

Report R02 — Radiomics and Overall Survival in Glioblastoma

A positive prognostic result, with the positive controls R01 lacked.

7 August 2026 • Quantara • Companion to R01 (MGMT classification, null)

Summary

Radiomic features from routine preoperative MRI **do** carry prognostic information about overall survival in UPENN-GBM: Harrell C-index **0.602** (95% CI 0.576–0.629), permutation $p = 0.005$, on 574 patients. The signal is *independent of age* — adding the radiomic score to age improves model fit substantially (likelihood-ratio $\chi^2_1 = 36.4$, $p = 1.6 \times 10^{-9}$), and the two together reach C-index 0.647.

Critically, both pre-registered positive controls recovered their known associations: older age predicts shorter survival ($p = 2.2 \times 10^{-18}$) and MGMT methylation predicts longer survival (median 595 vs 371 days, log-rank $p = 5.8 \times 10^{-8}$).

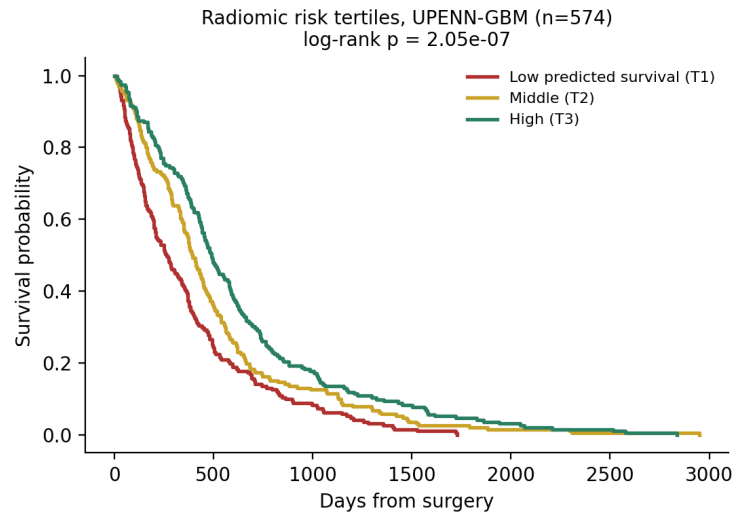
Read alongside R01, this produces a coherent and specific conclusion: these radiomic features encode prognosis but do *not* encode MGMT methylation status. The R01 null was therefore not a broken pipeline — the identical features and machinery find survival signal at $p = 0.005$ while finding nothing for MGMT.

1 Design

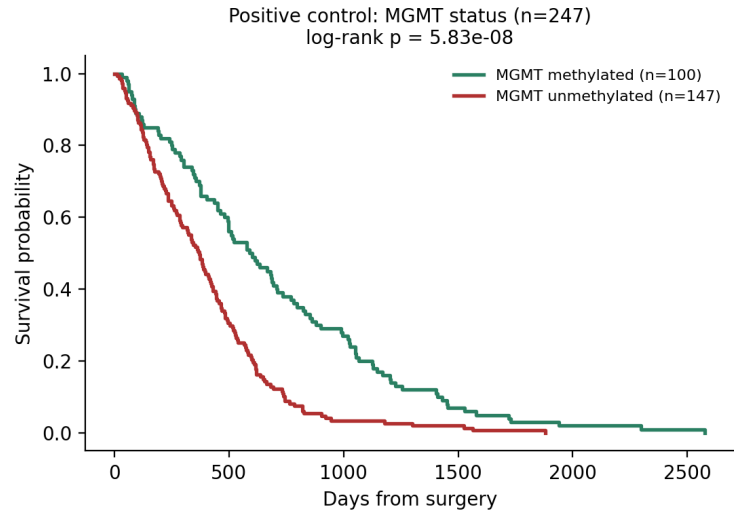
- **Cohort:** UPENN-GBM, 574 patients with complete features across all twelve modality-ROI files and a recorded survival time.
- **No censoring.** Every subject with a survival time in this collection is deceased (gate G3). Regression on $\log_{10}$ survival days is therefore valid without a Cox model for the primary. Median survival 388 days (IQR 173–636).
- **Features:** the same 1,728 CaPTk features used in R01 — four structural sequences $\times$ three regions $\times$ 144 descriptors.
- **Model:** elastic-net regression on log survival, $10\times$ repeated 5-fold nested CV, scaling and selection fitted inside training folds only.
- **Pre-specified signal rule:** prognostic only if the C-index 95% CI lower bound exceeds 0.55. Fixed in a config file hashed before any outcome was read.

2 Results

	Metric	Value	95% CI / p
Primary	Radiomics $\rightarrow$ survival, C-index	0.602	0.576–0.629
	Spearman (predicted vs observed)	0.302	$p = 1.3 \times 10^{-13}$
	Permutation null (200 permutations)	0.4998	observed $p = 0.005$
Control 1	Age $\rightarrow$ survival, C-index	0.622	0.594–0.647
	Spearman (age vs survival)	−0.354	$p = 2.2 \times 10^{-18}$
Control 2	MGMT methylated, median survival	595 days	log-rank $p = 5.8 \times 10^{-8}$
	MGMT unmethylated, median survival	371 days	
Joint	Age alone, Cox concordance	0.622	HR 1.35, $p = 5.8 \times 10^{-11}$
	Radiomics alone	0.602	HR 0.75, $p = 1.1 \times 10^{-10}$
	Age + radiomics	0.647	LR $\chi^2_1 = 36.4$, $p = 1.6 \times 10^{-9}$



Radiomic risk tertiles, log-rank $p = 2.0 \times 10^{-7}$.



Positive control. MGMT methylation recovers its established survival benefit.

3 Why this strengthens R01 rather than contradicting it

R01 reported that MGMT status is not predictable from these features (AUROC 0.430, 95% CI 0.360–0.500). A reasonable objection was that the pipeline might simply be broken. R02 answers that objection with evidence rather than assertion:

1. The same features and the same cross-validation machinery detect a survival signal at $p = 0.005$ against a correctly centred null.
2. MGMT status *is* biologically meaningful in this cohort — it separates survival at $p = 5.8 \times 10^{-8}$. So R01’s null is not because MGMT is irrelevant here.
3. Two independent known associations were recovered at the expected effect direction.

The specific conclusion is therefore narrow and defensible: **radiomic appearance tracks prognosis but does not track MGMT methylation**. Prognostic information in glioma imaging is not reducible to this particular molecular marker.

4 Limitations, stated plainly

- **No external validation is possible with our holdings.** The 337-patient CC0 cohort intended for external validation has no survival data at all. This result is internally validated only, and should not be presented otherwise.
- **Uncensored by exclusion.** Patients still alive (17) have no recorded survival time and are therefore absent. The cohort is effectively “time to death among those who died”, which removes censoring but introduces mild selection.
- **Effect size is modest.** C-index 0.602 alone is below age alone (0.622). The defensible claim is incremental value over age, not replacement of it.
- Automatic rather than manual segmentations were used, for sample size. Segmentation quality was not inspected.

5 Audit

Five checks were run after the result. Full record in `AUDIT.md`.

Check	Verdict	Finding
Is it just age?	No	Spearman(prediction, age) = -0.125 ; radiomics adds beyond age at $p = 1.6 \times 10^{-9}$
Convergence warnings	Immaterial	C-index identical (0.5978) at 5,000 / 50,000 / 200,000 iterations
Leakage	None	Permutation null centred 0.4998
Positive controls	Both recovered	Age and MGMT, correct directions, gate G6
Gates	7 / 7 PASS	Including complete-case rule identical to R01

6 How to verify

`evidence/oof_survival.csv` contains observed survival days and the predicted log-survival for all 574 patients, so the C-index is recomputable directly:

```
python3 -c "  
import numpy as np; from lifelines.utils import concordance_index  
d=np.loadtxt('evidence/oof_survival.csv',delimiter=',',skiprows=1)  
print(concordance_index(d[:,0], d[:,2]))" # -> 0.602
```

Also included: `config.yaml` (hash `f683cfed...`, written before outcomes were read), `gates.json`, `results.json`, `audit_permutation.json` (all 200 null values), `cohort.json`, `inputs.json`, `MANIFEST.sha256`, `env.txt`, and `m2_survival.py`.

7 Recommendation

This is the endpoint the glioma arm should be built on. It has ground truth in an imaged cohort, it replicates known biology as a control, and it yields a modest but genuine and age-independent signal.

Two things would materially strengthen it, both requests rather than computations: survival data for an independent imaged cohort (none of our holdings has it), and the UT Southwestern images, whose 625 patients span grades 2–4 and would allow the grade question the current cohorts cannot answer.